# MARCO BIELLA

**Researcher in Cognitive and Social Sciences / Data Scientist**

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## EXPERIENCE

**Postdoc Researcher**
Behavioral Marketing
University of Basel
February 2023 – Today

**Lecturer in Psychometrics**
Psychometrics
University of Bergamo
November 2024 – October 2025

**Postdoc Researcher**
Mobile App-based Attitude Change
Catholic University of the Sacred Heart – Milan
June 2022 – December 2022

**Postdoc Researcher**
Cognitive Research in Social Psychology
Heidelberg University (Prof. Klaus Fiedler)
October 2021 – September 2022

**Postdoc Researcher**
Machine Learning in Psychometrics, Neuroscience, and Neuroeconomics
University of Pisa
March 2021 – August 2021

**Postdoc Researcher**
Social Cognition and Decision Sciences
University of Tuebingen
March 2019 – February 2021

**Data Scientist**
Machine Learning and Predictive Analytics
Fault Detection for Banking Mobile Apps
October 2018 – December 2018

**Visiting Ph.D. Student**
Social Cognition Center Cologne
University of Cologne
November 2017 – March 2018

**Ph.D. Student**
Department of Psychology
University of Milano - Bicocca
November 2015 – October 2018 (former Lab Manager, March 2014 – February 2015)

**Online Media Intern**
Yahoo! Italia, Editorial Team & Online Communication
August 2011 – March 2012

## EDUCATION

**University of Milano - Bicocca**

| | |
|---|---|
| Ph.D. in Psychology, Linguistics, and Cognitive Neurosciences, *with honors* | 2015 – 2018 |
| M.Sc. Social Psychology and Economic Behavior, *cum laude* | 2012 – 2015 |
| B.A. Psychology of Communication | 2007 – 2011 |

# Scientific Production

## Publications

**Biella, M.**, & Batzdorfer, V. (2026). Editorial: The phenomenon of misinformation in different domains and by various disciplines. *Frontiers in Psychology*. 16:1763023. https://doi.org/10.3389/fpsyg.2025.1763023

**Biella, M.,** & Hütter, M. (2026). Navigating the Social Environment: A Sampling Approach to Trustworthiness. *Personality and Social Psychology Bulletin*. https://doi.org/10.1177/01461672241273237

**Biella, M.**, Gemignani, A., Conversano, C., Miniati, M., & Orrù, G. (2025). Psychometric Assessment of the Italian Version of the Vaccination Attitudes Examination (VAX) Scale and Exploration of its Link with Policy Endorsement. *Collabra: Psychology*. https://doi.org/10.1525/collabra.136895

**Biella, M.**, Hennig, M., & Oswald, L. (2025). Investigating the Social Boundaries of Fairness by Modeling Ultimatum Game Responders' Decisions with Multinomial Processing Tree Models. *Games*, *16*(1), 2 https://doi.org/10.3390/g16010002.

Anzani, A., **Biella, M.**, Prunas, A., Baldissarri, C., & Spaccatini, F. (2025). The call of the wolfpack. How social attraction toward the harasser complements ambivalent sexism in inducing empathic failure with the victim of stranger harassment. *Sexuality & Culture*. 1-21. https://doi.org/10.1007/s12119-025-10371-y

Rebholz, T., **Biella, M.,** & Hütter, M. (2024). Mixed-Effects Regression Weights for Advice Taking and Related Phenomena of Information Sampling and Utilization**.** *Journal of Behavioral Decision Making*. 37(2), https://doi.org/10.1002/bdm.2369

Pavanello Decaro, S., Pessina, R., **Biella, M.**, & Prunas, A. (2024). Italian women who have sex with women: prevalence and co-occurrence of sexual practices. *Sexual Medicine*. 12(2), https://doi.org/10.1093/sexmed/qfae017

**Biella, M.,** Rebholz, T., Holthausen, M., & Hütter, M. (2023). The Interaction Game: Development and Validation of an Experimental Paradigm for Manipulating Social Distance. *Journal of Applied Social Psychology*. 53(8), https://doi.org/10.1111/jasp.12969

**Biella, M.,** Orrù, G., Ciacchini, R., Conversano, C., Marazziti, D., & Gemignani A. (2023). Anti-vaccination attitude and vaccination intentions against COVID-19: a retrospective cross-sectional study investigating the role of media consumption. *Clinical Neuropsychiatry*, 20(4). https://doi.org/10.36131/cnfioritieditore20230404

Catellani, P., **Biella, M.,** Carfora, V., Nardone, A., Brischigiaro, L., Manera, M.R. & Piastra, M. (2023). A theory-based and data-driven approach to promoting physical activity through message-based interventions. *Frontiers in Psychology*. 14, https://doi.org/10.3389/fpsyg.2023.1200304

Frisanco, A., **Biella, M.,** Brambilla, M., & Kret, M. E. (2023). All that meets the eye: The contribution of reward processing and pupil mimicry on pupillary reactions to facial trustworthiness. *Current Psychology*. 42, https://doi.org/10.1007/s12144-021-02486-w

Carfora, V., **Biella, M.**, & Catellani, P. (2022). Affective Components in Promoting Physical Activity: A Randomized Controlled Trial of Message Framing. *Frontiers in Psychology*. 13, https://doi.org/10.3389/fpsyg.2022.968109

Anzani, A., **Biella, M.,** Scandurra, C., & Prunas, A. (2022). Desire for Genital Surgery in Trans Masculine Individuals: The Role of Internalized Transphobia, Transnormativity and Trans Positive Identity. *International Journal of Environmental Research and Public Health*. 19(15), https://doi.org/10.3390/ijerph19158916

Batzdorfer, V., Steinmetz, H., **Biella, M.**, & Alizadeh, M. (2021). Conspiracy theories on Twitter: emerging motifs and temporal dynamics during the COVID-19 pandemic. *International Journal of Data Science and Analytics*. 13, https://doi.org/10.1007/s41060-021-00298-6

Orrù, G., Bertelloni, D., Diolaiuti, F., Mucci, F., Di Giuseppe, M., **Biella, M.,** Gemignani, A., Ciacchini, R. & Conversano, C. (2021). Long-COVID Syndrome? A Study on the Persistence of Neurological, Psychological and Physiological Symptoms. *Healthcare*. https://doi.org/10.3390/healthcare9050575

Pagliaro, S., …, **Biella, M.**, et al. (2021). Trust predicts COVID-19 prescribed and discretionary behavioral intentions in 23 countries. *PLOS ONE*. 16(3), https://doi.org/10.1371/journal.pone.0248334

Brambilla, M., Masi, M., Mattavelli, S., & **Biella, M.** (2021). Faces and sounds becoming one: Cross-modal integration of facial and auditory cues in judging trustworthiness. *Social Cognition*, 39(3), https://doi.org/10.1521/soco.2021.39.3.315

Brambilla, M., **Biella, M.,** & Kret, M. E. (2019). The power of pupils in predicting conforming behavior. *Social Influence*, 14(2), https://doi.org/10.1080/15534510.2019.1637775

Brambilla, M., **Biella, M.,** & Kret, M. E. (2019). Looking into Your Eyes: Approach and Avoidance Responses Are Modulated by Pupillary Signals. *Cognition and Emotion*, 33(3), https://doi.org/10.1080/02699931.2018.1472554

**Biella, M.,** & Sacchi, S. (2018). Not fair but acceptable... for us! Group membership influences the tradeoff between equality and utility in the ultimatum game. *Journal of Experimental Social Psychology*, 77, https://doi.org/10.1016/j.jesp.2018.04.007

Brambilla, M., **Biella, M.,** & Freeman J. (2018). The Influence of Visual Context on the Evaluation of Facial Trustworthiness. *Journal of Experimental Social Psychology*. 78, https://doi.org/10.1016/j.jesp.2018.04.011 (The first two authors contributed equally to this article)

## Publications (working papers)

**Biella, M.,** Batzdorfer, V., Weitzel, A. I. & Unkelbach, C. (*in preparation*). Large Language Models' Alignment with Human Cognition. Exploring the Density Hypothesis in LLMs' Internal Representations and Generative Production.

Frisanco, A., Cantoni, C., Provenzano, L., **Biella, M.**, & Aglioti, S. M. (*under review, registered report*). Using Immersive Virtual Reality to Empower Women's Leadership: Embodying Female Role Model Avatars to Enhance Public Speaking. *The Leadership Quarterly*.

## Presentations

**Biella, M.** Information Sampling as a Two-Way Bridge Connecting Minds and the Environment. Organized Symposium at the *General Meeting of the European Association of Social Psychology 2026*, Strasbourg (France)

**Biella, M.** Information Sampling as a Two-Way Bridge Connecting Minds and the Environment. Organized Symposium at the *Fachgruppentagung Sozialpsychologie 2025*, Bochum (Germany)

**Biella, M.,** Hütter, M. Navigating the Social Environment Via a Sampling Approach to Trust. *Subjective Probability, Utility, and Decision Making 2025*, Lucca (Italy)

**Biella, M.** Hütter, M. Navigating the Social Environment Via a Sampling Approach to Trust. *First International Network on Trust 2025*, Genoa (Italy)

**Biella, M.** Big Data for Online Conversation Tracking. Scalable and Interpretable Topics from Online Discourse on Social Media. *SAMS 2024*, Luzern (Switzerland)

**Biella, M.** How agents' cognitive processes shape self-determined information search and the resulting judgments and decisions. *TeaP 2023*, Trier (Germany)

**Biella, M.**. Social Environment and Information Structure Effects. *ESCON 2022*, Milan (Italy)

**Biella, M.,** & Hütter, M. Goals Affect Impression Biases Via Information Search Strategies and Sampling Behaviors. *13th Annual General Meeting of the Society for the Science of Motivation 2021*.

**Biella, M.,** & Hütter, M. Navigating the Social Environment: A Sampling Approach to Trustworthiness. *EASP General Meeting 2020*, Krakow (Poland) (COVID19)

**Biella, M.,** & Hütter, M. The Interaction Game: Development and Validation of an Experimental Paradigm for Manipulating Social Distance. *TeaP 2020*, Jena (Germany)(COVID19)

**Biella, M.,** & Hütter, M. Navigating the Social Environment: A Sampling Approach to Trustworthiness. *ESCON 2019*, Bordeaux (France)

**Biella, M.,** Speckmann, F., & Brambilla, M. Rearranging the Face Space, an Exploration using the Spatial Arrangement Method. *ESCON 2018*, Cologne (Germany)

**Biella, M.,** Zogmaister, C., Ceolato, S., & Parozzi, E. Twitter Twitter on the wall, which university's the fairest of them all? *Big Data in Psychology 2018*, Trier (Germany)

**Biella, M.,** Zogmaister, C., Ceolato, S., & Parozzi, E. Twitter Twitter on the wall, which university's the fairest of them all? *Big Data, Big Challenges. Mid-term Conference, Italian Association of Sociology, Methodology Section 2016*, Milan (Italy)

**Biella, M.,** & Sacchi, S. Not fair but acceptable... for us! Group membership influences the tradeoff between equity and utility in the Ultimatum Game. *ESCON 2016*, Lisbon (Portugal)

## Awards

**Top Cited Article 2025**. Journal of Behavioral Decision-making. *Rebholz, Biella & Hütter (2024)*
**Best Social Psychology Paper 2019**. Italian Association of Psychology. *Biella & Sacchi (2018)*
**Best Doctoral Dissertation 2019**. Italian Association of Psychology.

## Funding

WWZ-Forum 2023. Aggressive Repositioning via Celebrity Endorsement in Advertising, With Prof. Miguel Brendl (27787 CHF)

Google Cloud for Researchers 2022. PI: Dr. Marco Biella. Topic Modeling on the Discourse around Political Events on Twitter using Big Data and Latent Dirichlet Allocation (€ 4666)

EASP Seedcorn Grant 2020. PI: Dr. Marco Biella. Far from fair: Social closeness affects the tradeoff between utility and fairness in decision-making (€ 2340)

EASP Travel Grant 2018. European Association of Social Psychology Travel Grant - EASP Summer School 2018, Workshop on Motivation and control (€ 615)

FAR 2017. PI: Prof. Marco Brambilla. Split-second detectability of moral characteristics from faces: Determinants and social implications (€ 2271)

FAR 2015 - 2016. PI: Prof. Marco Brambilla. Your pain makes me laugh: Psychological consequences of morally wrong sources of pleasure (€ 6561)

# Teaching

## Teaching Experience

2025 Lecturer. *Behavioral Marketing*. University of Basel.
2024 Lecturer. *Psychometrics*. University of Bergamo.
2025, 2024, 2023 Teaching staff. *Behavioral Science*. University of Basel.
2025, 2024, 2023 Teaching staff *Consumer Behavior: Theoretical Foundations*. University of Basel.
2025, 2024, 2023 Teaching staff. *Psychological Theories in Consumer Behavior*. University of Basel.
2024, 2023 Teaching staff. *Behavioral Marketing*. University of Basel.
2021 Lecturer. *Practical Exploratory Data Analysis and Data Management*. Heidelberg University.
2020, 2019 Lecturer. *Social Psychology*. Eberhard Karls University of Tübingen
2018 Teaching staff. *Psychometric Methods*. University of Milano–Bicocca.
2017 Teaching staff. *Social Psychology of Groups*. University of Milano–Bicocca.

## Thesis Supervisions

Samuel Wilson M.Sc. 5.4 (Basel[i])
Yorick Kirnbauer M.Sc. 5.9 (Basel[i])
Sewin Therampilly M.Sc. 5.6 (Basel[i])
Philipp Behringer B.Sc. 5.8 (Basel[i])
Jakob Winterhalder M.Sc. 5.8 (Basel[i])
Brais Canosa Calo M.Sc. 5.5 (Basel[i])
Dilan Tasyurdu M.Sc. 4.3 (Basel[i])
Laura Oswald M.Sc., 1.0 (Tübingen[ii])
Julia Henner B.Sc., 1.0 (Tübingen[ii])
Adriana Kantorova B.Sc., 1.3 (Tübingen[ii])
Matteo Masi M.Sc., 110 *cum laude* (Milan[iii]) *cosupervision
Althea Frisanco M.Sc., 110 *cum laude* (Milan[iii]) *cosupervision
Beatrice Leonardi M.Sc., 110 *cum laude* (Milan[iii]) *cosupervision
Fiorella Carminati M.Sc., 110 *cum laude* (Milan[iii]) *cosupervision

# Misc

## Professional Affiliations / Habilitation

Qualification aux fonctions de Maître de Conférences (section 16 - psychologie)
Society for Judgment and Decision Making
Responsible Digital Society Research Network
EASP, European Association of Social Psychology
AIP, Italian Association of Psychology – Experimental Psychology
AIP, Italian Association of Psychology – Social Psychology

## Conference Organization

International Small Group Meeting - Social Cognition and Decision Sciences. July 4th – 7th, 2019, Tübingen University (Conference Funded by DFG Grant awarded to Prof. Mandy Hütter)

## Service to the Scientific Community

Frontiers in Psychology, Topic Editor - Misinformation
Invited to the EASP Vision Workshop, 16-17 Oct. 2019, Amsterdam, The Netherlands
Peer reviewer (Behavior Research Methods, Scientific Reports, PLOS ONE, Group Processes and Intergroup Relations, Frontiers in Psychology, Acta Psychologica, Social Influence)

# Competencies

## Certifications

- Intro to Time Series Analysis in R (2021)
- Gentle Introduction to Bayesian Statistics (2018)
- Practical Machine Learning (2018)
- Regression Models (2017)
- Reproducible Research (2017)
- Statistical Inference (2017)
- Exploratory Data Analysis (2017)
- Getting and Cleaning Data (2016)
- R Programming (2016)
- The Data Scientist's Toolbox (2016)
- Meta-analysis and Results Replicability (2016)

## Research Methods, Data Science

- Machine Learning (xgboost, random forest, svm, regularized regression, naive bayes, neural networks)
- Explainable Artificial Intelligence XAI (LIME, Shapley Values)
- Natural Language Processing (NLP), Topic Modeling with Latent Dirichlet Allocation
- Large Language Models / AI
- Bayesian Statistics
- Linear Mixed-Models
- Structural Equation Modeling
- Psychometrics
- Network Analysis
- Cluster Analysis (hierarchical, kmeans, dbscan)
- Multinomial Processing Tree Models
- Fault Detection, Isolation Forest

## Programming Language / Software

R, Python, Tensorflow, Pytorch, cloud computing and virtual machines, Google Cloud Platform, Linux and Unix Systems, on-premise computing infrastructure (e.g., university-managed computing clusters), APIs, git.

## Large Language Models / AI

Proprietary and Open-source models (Google Gemini, Meta Llama, DeepSeek R1 and distilled versions, etc.), APIs for hosted models (Google Gemini, together.ai), locally-hosted models (LM Studio, Ollama), huggingface ecosystem and transformer library.

## Research Tools

PsychoPy/Pavlovia, openSesame, Gorilla, inquisit, Qualtrics, Prolific, Cloudresearch.

# Reference Contacts

Prof. Klaus Fiedler, Heidelberg University - klaus.fiedler@psychologie.uni-heidelberg.de.
Prof. Christian Unkelbach, Social Cognition Center Cologne - christian.unkelbach@uni-koeln.de.
Prof. Mandy Hütter, University of Tuebingen - mandy.huetter@uni-tuebingen.de.
Prof. Rainer Greifeneder, University of Basel - rainer.greifeneder@unibas.ch.
Prof. Antonio Loprieno, University of Basel - a.loprieno@unibas.ch.

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[i] Swiss grading system: from 1 (lowest) to 6 (highest).
[ii] German grading system: from 5 (lowest) to 1 (highest).
[iii] Italian grading system: from 0 (lowest) to 110 (highest).